

Q1. Write programs according to given description:

a. Write a program to input value of x. Further write code to evaluate and print the expression: $\frac{3x^3 - 5x^2}{2x}$

b. Compute and display value of **P** from the following equation. Interactive code should input required values from user.

$$P = \frac{s + (2+r)^n}{1-r^n}$$

c. Write print statements of C program to produce exact output:

Expression: "b x c", where b = 5 & c = 4. Answer was 20.
'\t' is a special character.

```
int a=3, b =5, c=4; //use variables to print numbers in output
```

d.

```
int main( ){           //After input statement, write code to print first value is greater or second
    int f = 1, s = 1; // value is greater or both are equal, no output if both values are zero
    while ( ) {        //Write appropriate condition inside while
        printf ("Enter two values with spaces (both 0 to stop): ");
        scanf("%d %d", &f, &s);
```

e.

```
    }
    i = 1;
    while (i<=10){
        printf ("%d ", i);
        if ( )
            i++;
    }
```

**//Write missing statements to print output as in the box
 //without modifying the given code**

1 2 : 3 4 : 5 6 : 7 8 : 9 10

f. Write a program that has three integers variable. The program should print 1 if the first number raised to the power of the second number equals the third number i.e. $a^b=c$

g. Fill table with values that are assigned to each variable after the execution of lines 3-6 by given code.

1. `int a = 3, b = 5, c = 14, sum = 0;`
2. `sum = a + b + c;`
3. `c = c / a;`
4. `b = b + c - a++;`
5. `a = 2 * --b + c;`
6. `sum = a + b % c * c - b;`

Line No.	a	b	c	sum
Line 3				
Line 4				
Line 5				
Line 6				

h. A company delivers snacks in packs only. The pack size is eight snacks per pack. They received order of snacks in bulk. However, they calculate the number of packs required. If the last pack has snacks less than eight snacks, they deliver a full pack without extra charges of extra snacks. You have to write program for input three order of snacks and calculate extra snacks in each order and total extra snacks in three orders. See sample run for your understanding:

Sample Run:
 Order 1: 85
 Order 2: 122
 Order 3: 166
 Extra Snacks 1: 3
 Extra Snacks 2: 6
 Extra Snacks 3: 2
 Total Extra Snacks: 11

- i. Write a program to input two numbers. If first number is smaller than second number, print integers from first number to second number, excluding second number. If first number is equal or larger then give appropriate message(s). See sample run for your understanding:

Sample Run:

```
N1: 20
N2: 20
N1 = N2
N1: 20
N2: 18
N1 > N2
N1: 20
N2: 24
20 21 22 23
```

- j. Write a program-using while loop. A student has appeared in the exams. There are total 10 subjects. He has failed to appear in some subjects. Input his marks in the subject, only if he has appeared in the exam and find average marks in 10 subjects and average marks in the subjects, he has appeared in the exam. See sample run for your understanding:

Sample Run:

```
Exam 1: Y
Marks: 85
Exam 2: Y
Marks: 75
Exam 3: N
Exam 4: Y
...
Average Marks: 44
Average Marks in
appeared exams: 65
```

- k. Write a program using while loop to print output (given inside the box).

```
M *
N #
O *
P #
```

Q2. Show complete working of following codes and print output in single line:

- a.

```
a = b = c = 2;
while (b < 20){
    printf ("%d ", a);
    a = c - b;
    b = b + 1;
    c = c + 2;
}
```
- b.

```
x = 5;
while (x > 0){
    printf ("%d ", x);
    t = t - x;
    x = x - 2;
}
printf ("%d ", t);
```
- c.

```
x = 8, y = 6, z = 5;
if (x > y){
    x = x + y;
    y = x - y;
    x = x - y;
}
if (y > z){
    temp = y;
    z = y ;
    y = temp;
}
printf ("%d %d %d", x, y, z);
```

Q3. Read code and requirement. See output carefully and explain any logical error exist in the code:

Note: You are not required to correct the code. If you try to write the correct code, it has no consideration.

- a. Print integers between 1-99 divisible by 3 and 5:

```
i = 1;
while (i < 100){
    if (i % 3 == 0)
        printf ("%d ", i);
    if (i % 5 == 0)
        printf ("%d ", i);
    i = i + 1;
}
```

3 5 6 9 10 12 15 18 ... 99

- b. Print 20 integers between 1-99 divisible by 2 and 3:

```
i = 1;
while (i <= 10)
    if (i % 2 == 0 && i % 3 == 0){
        printf ("%d ", i);
        i = i + 1;
    }
```

6 12 18 24 30 36 42 48 54 60

- c. Print 20 random alphabets including both capital and small:

```
i = 1;
while (i <= 20){
    c = rand() % 57 + 65;
    printf ("%c ", c) ;
    i = i + 1;
}
```

#122 is ASCII code of z and 65 is ASCII code of A